Duty at home

DM13 · Reservations and overbooking · 60 min.

A room has eighteen places. The organiser accepts twenty reservations, paid in advance.
Each person who has booked presents himself with a probability of 0 ,8 independently of the others.
Note X If more than 18 people come, the number of people who come to see each other will increase.
surplus persons are refused.

1. Justifying the law of Xthen calculate his expectation and variance.

2. Calculate the probability that the organizer should refuse at least one person. -6.

3. Each reservation returns 10 monetary units, including if the person does not come.
   person refused receives a total payment of 30 monetary units, reimbursement and compensation
   Understood. $R = \max(X - 18; 0)$ the number of people refused. Express the gain G in
   function of R, then calculate $E(G)$.

4. Calculate the gain if the organiser accepts only eighteen reservations, with the same rules of
   Payment. Compare the two strategies from the point of view of average gain.

5. Citing a model hypothesis that can be questionable in practice, then a criterion other than the
   monetary gain to be taken into account in this decision.